

Complexity	Dataset	Images	Modality	Prompt	TotalSegmentator	BiomeParse	Text3DSam	SegVol	BiomedParseV2	SAT	VoxTel
Healthy (Known Concept)	Pediatric-CT-SEG [84]	21	CT	gallbladder	78.91	8.74	7.25	71.51	84.63	80.25	85.28
				duodenum	53.16	2.68	6.47	51.56	59.05	62.17	66.97
				right adrenal gland	41.27	0.38	0.01	37.99	40.52	46.84	48.52
				left adrenal gland	42.53	1.17	2.01	45.27	44.32	52.71	52.57
				urinary bladder	83.45	38.17	35.63	81.41	81.47	82.59	79.96
				bones	90.21	0.74	2.35	83.8	0.01	8.64	93.62
				left breast	23.41	0	0	0	0	39.86	41.52
				right breast	24.01	0	0	0	0	39.91	45.07
				esophagus	61.67	8.17	25.46	59.11	64.16	66.48	65.96
				heart	81.99	2.9	68.02	4.49	85.69	90.29	48.95
				left kidney	94.17	22.86	59.28	93.05	96.22	96.12	95.31
				right kidney	94.45	18.77	28.35	92.17	94.36	94.26	94.7
				large intestine	69.54	3.25	7.29	6.52	5.69	11.98	63.88
				liver	95.61	35.62	76.64	95.76	96.28	95.65	95.56
				left lung	86.38	0	77.16	87.23	80.49	87.73	87.65
				right lung	86.13	0	79.83	89.77	84.57	88.22	87.65
				pancreas	78.1	11.66	14.3	72.62	78.68	78.81	77.6
				rectum	0	0.69	0	25.99	0	68.76	48.24
				skin	41.44	0.01	0	5.45	0	69.11	63.16
				small intestine	72.49	3.15	8.29	28.60	23.46	71.13	70.00
	AeroPath [186]	27	CT	lung	96.44	0	68.14	97.28	54.27	96.93	96.99
				trachea	74.02	0.01	62.58	80.06	70.92	79.02	82.31
	VEELA2025 [189]	20	CT	liver	95.19	17.01	77.88	94.25	94.67	94.6	94.47
				hepatic vessels	37.77	2.59	5.81	29.76	6.76	9.37	41.17
	RADCURE [210]	22	CT	portal vessels	15.49	0	3.05	0	3.75	8.99	46.04
				brainstem	-	3.22	2.12	69.61	65.12	82.57	82.27
				left cochlea	-	0	0	0	0	18.15	19.58
				right cochlea	-	0	0	0	0	21.49	23.31
				esophagus	-	25.98	12.51	40.42	55.36	63.71	61.63
				left eye	-	1.73	0	40.64	11.54	87.47	85.2
				right eye	-	1	0	28.31	3.72	86.14	83.12
				left lacrimal gland	-	0	0	0	9.99	28.07	0
				right lacrimal gland	-	0.01	0	0.06	18.36	32.07	0
				left submandibular gland	-	0.71	0.71	27.91	59.66	76.65	77.14
				right submandibular gland	-	0.88	0.43	31.89	59.02	71.91	71.47
				larynx	-	0.72	0	2.79	17.15	47.14	39.55
				left lens	-	0.07	0	1.19	0	62.59	74.32
				right lens	-	0.03	0	0.88	0	63.66	74.74
				lips	-	0	0.23	20.8	13.05	26.62	25.39
				mandible	-	0.07	0.03	82.83	0	86.87	91.55
				superior constrictor muscle	-	0.79	0	0	0.16	32.95	3.37
				middle constrictor muscle	-	0.94	0	0	0	18.63	1.73
				inferior constrictor muscle	-	2.8	0.01	0	0.04	1.56	5.11
				left optic nerve	-	0.39	0.58	5.49	1.03	63.41	66.58
				right optic nerve	-	0.09	0.13	14.3	4.97	58.08	67.32
				chiasma opticum	-	0	0	0	0	39.49	40.14
				left parotid gland	-	0.12	1.72	1.14	59.85	82.84	83.51
				right parotid gland	-	0.15	0.04	5.03	54.75	82.9	83.71
				spinal cord	-	0.11	13.74	56.98	61.9	68.91	69.87
	HVSMR-2.0 [154]	60	CMR	left ventricle	-	26.63	34.9	0.05	17.8	53.54	46.91
				right ventricle	-	16.21	18.96	5.01	10.42	43.39	55.12
				left atrium	-	12.12	26.32	0	20.49	47.31	68.23
				right atrium	-	6.04	21.19	0.02	10.74	47.85	50.39
				aorta	-	1.31	5.9	5.76	44.78	53.71	70.68
				pulmonary artery	-	1.5	4.51	0	12.73	39.73	48.43
				superior vena cava	-	0.2	4.23	0.32	4.88	36.19	47.63
				inferior vena cava	-	0.93	9.18	3.63	23.03	25.3	34.4
Pathological (Known Concept)	ARCIN NSCLC [94]	44	PET	fdg-avid tumor	-	2.73	12.21	0	0.47	77.13	83.24
	ISBI MS [37]	42	MR (FLAIR)	MS lesion	-	9.57	0.21	0	2.03	13.68	72.71
	Adrenal-ACC-Ki67 [148]	52	CT	tumor	-	52.91	0	51.08	55.86	0.09	77.23
	HCC-TACE-SEG [147]	65	CT	hepatocellular carcinoma	-	41.25	0.52	58.35	70.37	62.28	73.24
	Penguin [129]	100	CT	sacrum	-	1.05	15.24	64.55	3.21	95.33	96.48
				left hipbone	-	0.14	33.93	0.02	0.69	96.45	98.24
				right hipbone	-	0.21	25.09	1.75	0.15	96.37	98.05
Cross-Modality	BrainMetShare [66]	105	MR (T1)	lesion	-	9.61	0.1	0	18.66	22.16	48.19
	QIN Breast [118]	36	PET	fdg-avid breast cancer	-	0.75	0	0.05	0	58.26	72.27
	PANTHER Task 1 [27]	92	MR (T1)	pancreas tumor	-	1.39	4.52	14.64	18.24	19.25	35.66
	Soft-Tissue-Sarcoma [191]	102	MR (T1/T2)	soft tissue lesion	-	36.20	12.83	0.83	7.68	10.64	40.34
Unknown Concept	FedBCa (Center2) [36]	47	MR (T2)	carcinoma	-	10.37	2.04	0.08	2.69	9.56	25.76
	MedSeg Esophageal [1]	1	CT	esophageal cancer	-	17.86	7.28	32.24	16.56	0	69.07

Table 7. **Overview of held-out test datasets.** This table presents the full set of evaluation results across all datasets, including healthy and pathological known concepts, as well as generalization to unseen modalities and classes. Dataset metadata is also included. For partially annotated datasets such as Pediatric-CT-SEG and RADCURE-Structures, only the subset containing all annotated classes were used.

Region	Subregion	Structures (Count)
Brain	Cerebral structures	brainstem (7360), brain (4330), left cerebral white matter (3654), right cerebral white matter (3654), left cerebral cortex (3644), right cerebral cortex (3644), white matter (519), gray matter (489), white matter hyperintensities (60), deep gray matter (40), cortical gray matter (30), white matter lesions (30)
	Ventricles and csf	cerebrospinal fluid (4133), left lateral ventricle (3654), right lateral ventricle (3654), left inferior lateral ventricle (3644), right inferior lateral ventricle (3644), 3rd ventricle (3644), 4th ventricle (3644), left brain lateral ventricle (158), right brain lateral ventricle (158), lateral ventricles (80), third ventricle (10), fourth ventricle (10)
	Basal ganglia	left caudate (3654), right caudate (3654), left putamen (3654), right putamen (3654), left pallidum (3654), right pallidum (3654), left accumbens area (3644), right accumbens area (3644), left caudate nucleus (158), right caudate nucleus (158), left lentiform nucleus (158), right lentiform nucleus (158), basal ganglia (30), left nucleus accumbens (10), right nucleus accumbens (10)
	Thalamus and diencephalon	left thalamus (3654), right thalamus (3654), left ventral dc (3644), right ventral dc (3644), left ventral diencephalon (10), right ventral diencephalon (10)
	Hippocampus	left hippocampus (3932), right hippocampus (3932), anterior hippocampus (260), posterior hippocampus (260), left hippocampus subiculum (20), left hippocampus cornu ammonis 2 (20), left hippocampus cornu ammonis 1 (20), left hippocampus cornu ammonis 4 and gyrus dentatus (20), left hippocampus entorhinal cortex (20), left hippocampus cornu ammonis 3 (20), left hippocampus tail (20), right hippocampus subiculum (20), right hippocampus cornu ammonis 2 (20), right hippocampus cornu ammonis 1 (20), right hippocampus cornu ammonis 4 and gyrus dentatus (20), right hippocampus entorhinal cortex (20), right hippocampus cornu ammonis 3 (20), right hippocampus tail (20)
	Amygdala and limbic	left amygdala (3654), right amygdala (3654), corpus callosum (10)
	Cerebellum	left cerebellum white matter (3644), right cerebellum white matter (3644), left cerebellum cortex (3644), right cerebellum cortex (3644), cerebellum (70), cerebellar vermis superior posterior (10), cerebellar vermis anterior (10), cerebellar vermis inferior posterior (10), left cerebellar hemisphere (10), right cerebellar hemisphere (10)
	Frontal lobe	left caudal anterior cingulate (10), left caudal middle frontal (10), left lateral orbitofrontal (10), left medial orbitofrontal (10), left paracentral lobule (10), left pars opercularis (10), left pars orbitalis (10), left pars triangularis (10), left precentral gyrus (10), left rostral anterior cingulate (10), left rostral middle frontal (10), left superior frontal (10), left frontal pole (10), right caudal anterior cingulate (10), right caudal middle frontal (10), right lateral orbitofrontal (10), right medial orbitofrontal (10), right paracentral lobule (10), right pars opercularis (10), right pars orbitalis (10), right pars triangularis (10), right precentral gyrus (10), right rostral anterior cingulate (10), right rostral middle frontal (10), right superior frontal (10), right frontal pole (10)
	Temporal lobe	left temporal lobe (120), right temporal lobe (120), left temporal lobes (50), right temporal lobes (50), left banks superior temporal sulcus (10), left entorhinal (10), left fusiform gyrus (10), left inferior temporal (10), left middle temporal (10), left parahippocampal gyrus (10), left superior temporal (10), left temporal pole (10), left transverse temporal (10), right banks superior temporal sulcus (10), right entorhinal (10), right fusiform gyrus (10), right inferior temporal (10), right middle temporal (10), right parahippocampal gyrus (10), right superior temporal (10), right temporal pole (10), right transverse temporal (10)

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Region	Subregion	Structures (Count)
	Parietal lobe	left inferior parietal (10), left postcentral gyrus (10), left posterior cingulate (10), left precuneus (10), left superior parietal (10), left supramarginal gyrus (10), right inferior parietal (10), right postcentral gyrus (10), right posterior cingulate (10), right precuneus (10), right superior parietal (10), right supramarginal gyrus (10)
	Occipital lobe	left cuneus (10), left lateral occipital (10), left lingual gyrus (10), left pericalcarine (10), right cuneus (10), right lateral occipital (10), right lingual gyrus (10), right pericalcarine (10)
	Insula and cingulate	left isthmus cingulate (10), left insula (10), right isthmus cingulate (10), right insula (10)
	Other cortical	extra-axial space (40), left unknown cortex (10), right unknown cortex (10)
	Anterior circulation arteries	right internal carotid artery (833), left internal carotid artery (833), anterior communicating artery (300), right anterior cerebral artery (250), left anterior cerebral artery (250), right middle cerebral artery (250), left middle cerebral artery (250), third segment of anterior cerebral artery (a2) (250), right a1 and a2 of anterior cerebral artery (50), left a1 and a2 of anterior cerebral artery (50), right a3 of anterior cerebral artery (50), left a3 of anterior cerebral artery (50), third a2 of anterior cerebral artery (50), third a3 of anterior cerebral artery (50), right m1 of middle cerebral artery (50), left m1 of middle cerebral artery (50), right m2 of middle cerebral artery (50), right m3 of middle cerebral artery (50), left m2 of middle cerebral artery (50), left m3 of middle cerebral artery (50), paraophthalmic part of right internal carotid artery (1), posterior communicating part of right internal carotid artery (1), anterior choroidal part of the right internal carotid artery (1), right carotid terminus (1), right m1 (1), right a1 (1), anterior communicating artery partially fenestrated (1), right a2 (1), left paraophthalmic part of the internal carotid (1), left posterior communicating part of the internal carotid (1), left anterior choroidal artery part of the internal carotid (1), left carotid terminus (1), left a1 (1), left a2 (1), left m1 (1), insular segment of the right middle cerebral artery (1), insular branches of the left middle cerebral artery (m3/m4) (1), right anterior insular part of middle cerebral artery (m3/m4) (1), right posterior insular part of the middle cerebral artery (1), right superior terminal branch of the middle cerebral artery (1), right parietal cortical branches of the insular part of the middle cerebral artery (1), left m2-branches (1), left m3/m4 (1)
	Posterior circulation arteries	basilar artery (302), right posterior cerebral artery (250), left posterior cerebral artery (250), right superior cerebellar artery (51), left superior cerebellar artery (51), right vertebral artery (50), left vertebral artery (50), right p1 and p2 of posterior cerebral artery (50), left p1 and p2 of posterior cerebral artery (50), right p3 and p4 of posterior cerebral artery (50), left p3 and p4 of posterior cerebral artery (50), right anterior inferior cerebellar artery (50), left anterior inferior cerebellar artery (50), right posterior inferior cerebellar artery (50), left posterior inferior cerebellar artery (50), right p1 (1), right p2 (1), left p1 (1), left p2 (1), left p3 (1), left p4/p5 (1)
	Circle of willis	right posterior communicating artery (301), left posterior communicating artery (301), right anterior choroidal artery (51), left anterior choroidal artery (51)

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Region	Subregion	Structures (Count)
	Small cerebral arteries	right ophthalmic artery (50), left ophthalmic artery (50), left ophthalmic artery (1), right ophthalmic artery (1), right lenticulostriate arteries (1), left lenticulostriate arteries (1), right posteromedial central arteries (1), left posteromedial central arteries (1), right paramedian artery branch (1), right medial frontobasal artery (1), right lateral frontobasal artery (1), left medial orbitofrontal artery (1), left polar frontal artery (1), right polar frontal artery (1), left pericallosal artery (1), left callosomarginal artery (a3/4) (1), right callosomarginal artery (a3/4) (1), right anterior internal frontal artery (1), left anterior internal frontal artery (1), right intermediate internal frontal artery (1), left intermediate internal frontal artery (1), right prefrontal artery (1), right artery of precentral gyrus (1), right artery of central sulcus (1), right polar temporal artery (1), right temporal branch (1), right anterior temporal artery branch (1), right middle temporal artery branch (1), right posterior parietal artery (1), right posterior temporal artery branch of the mca (1), right parietal branches of the middle cerebral artery (1), right posterior temporal branches of the pca (1), right medial occipital artery (1), right precuneal branches (1), right calcarine branches (1), right parietooccipital branches (1)
	Cerebral veins and sinuses	superior sagittal sinus (26), vein of galen (25), straight sinus (25), internal cerebral veins (25), right basal vein of rosenthal (25), left basal vein of rosenthal (25)
	Meningeal vessels	right middle meningeal artery (26), left middle meningeal artery (26)
	Brain pathology	enhancing tumor (4681), edema (2840), necrosis/non-enhancing tumor core (2183), non-enhancing tumor core (1943), surrounding non-enhancing hyperintensity (1943), resection cavity (1943), brain lesion (655), gross tumor volume (500), stroke (453), multiple sclerosis lesion (366), necrotic tumor (294), non-enhancing tumor (261), cystic component (261), brain tumor (186), hie lesion (85), cerebral microbleed (72), infarction (30)
Head and neck	Eyes and orbits	left eye (3505), right eye (3505), left lens (3505), right lens (3505), left eyeball (533), right eyeball (533), eyeball globe (400), orbital fat (400), left anterior eyeball (84), right anterior eyeball (84), left posterior eyeball (84), right posterior eyeball (84)
	Extraocular muscles	eye superior rectus muscle (400), eye lateral rectus muscle (400), eye medial rectus muscle (400), eye inferior rectus muscle (400), eye superior oblique muscle (400)
	Optic pathway	left optic nerve (3636), right optic nerve (3636), chiasma opticum (3636), optic nerve (400)
	Lacrimal system	left lacrimal gland (3419), right lacrimal gland (3419), lacrimal gland (400)
	Facial structures	left cheek (533), right cheek (533)
	Inner ear	left cochlea (3539), right cochlea (3539), cochlea (210), left vestibular semicircular canal (120), right vestibular semicircular canal (120), left inner ear (50), right inner ear (50)
	Middle ear	left middle ear (170), right middle ear (170), left tympanic cavity (120), right tympanic cavity (120), left eustachian tube bone (120), right eustachian tube bone (120)
	Temporal bone	left internal auditory canal (120), right internal auditory canal (120), left mastoid process (120), right mastoid process (120)
	Cranial bones	mandible (3466), skull (3358), lower jawbone (532), upper jawbone (532), left inferior alveolar canal (532), right inferior alveolar canal (532), left mandibular incisive canal (532), right mandibular incisive canal (532), lingual canal (532), left mandible (170), right mandible (170), left temporomandibular joint (170), right temporomandibular joint (170)

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Region	Subregion	Structures (Count)
	Paranasal sinuses	nasal cavity (533), left maxillary sinus (532), right maxillary sinus (532)
	Upper teeth	upper right central incisor (532), upper right lateral incisor (532), upper right canine (532), upper right first premolar (532), upper right second premolar (532), upper right first molar (532), upper right second molar (532), upper right third molar (wisdom tooth) (532), upper left central incisor (532), upper left lateral incisor (532), upper left canine (532), upper left first premolar (532), upper left second premolar (532), upper left first molar (532), upper left second molar (532), upper left third molar (wisdom tooth) (532)
	Lower teeth	lower left central incisor (532), lower left lateral incisor (532), lower left canine (532), lower left first premolar (532), lower left second premolar (532), lower left first molar (532), lower left second molar (532), lower left third molar (wisdom tooth) (532), lower right central incisor (532), lower right lateral incisor (532), lower right canine (532), lower right first premolar (532), lower right second premolar (532), lower right first molar (532), lower right second molar (532), lower right third molar (wisdom tooth) (532)
	Tooth pulp	upper right central incisor pulp (532), upper right lateral incisor pulp (532), upper right canine pulp (532), upper right first premolar pulp (532), upper right second premolar pulp (532), upper right first molar pulp (532), upper right second molar pulp (532), upper right third molar (wisdom tooth) pulp (532), upper left central incisor pulp (532), upper left lateral incisor pulp (532), upper left canine pulp (532), upper left first premolar pulp (532), upper left second premolar pulp (532), upper left first molar pulp (532), upper left second molar pulp (532), upper left third molar (wisdom tooth) pulp (532), lower left central incisor pulp (532), lower left lateral incisor pulp (532), lower left canine pulp (532), lower left first premolar pulp (532), lower left second premolar pulp (532), lower left first molar pulp (532), lower left second molar pulp (532), lower left third molar (wisdom tooth) pulp (532), lower right central incisor pulp (532), lower right lateral incisor pulp (532), lower right canine pulp (532), lower right first premolar pulp (532), lower right second premolar pulp (532), lower right first molar pulp (532), lower right second molar pulp (532), lower right third molar (wisdom tooth) pulp (532)
	Dental prosthetics	bridge (532), crown (532), implant (532)
	Oral soft tissues	lips (3419), oral cavity (204), buccal mucosa (84)
	Salivary glands	left parotid gland (3636), right parotid gland (3636), left submandibular gland (3586), right submandibular gland (3586)
	Thyroid and pituitary	left thyroid (2226), right thyroid (2226), thyroid gland (1446), pituitary gland (254)
	Pharynx	pharynx (532), pharyngeal constrictor muscle (120), nasopharynx cancer (50)
	Larynx	larynx (3455), larynx glottis (204), larynx supraglottic (204), arytenoid cartilage (84), cricopharyngeal inlet (84)
	Carotid system	left common carotid artery (1931), right common carotid artery (1761), left carotid artery (98), right carotid artery (98), right external carotid artery (25), left external carotid artery (25), right superficial temporal artery (25), left superficial temporal artery (25), right maxillary artery (25), left maxillary artery (25), common carotid artery (1), internal carotid artery (1), external carotid artery (1)
	Jugular venous system	right internal jugular vein (533), left internal jugular vein (533), internal jugular vein (1), external jugular vein (1), posterior auricular vein (1), retromandibular vein (posterior branch) (1), facial vein (1), transverse cervical veins (1), superficial temporal veins (1), middle temporal vein (1), deep superficial veins (1), retromandibular vein (anterior branch) (1)
	Subclavian system	brachiocephalic vein (1), subclavian vein (1)

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Region	Subregion	Structures (Count)
	Other neck structures	head neck lymph node (245)
	Head neck pathology	head neck primary tumor (674), head neck metastatic lymph nodes (674), head neck lesion (245), vestibular schwannoma (210)
Chest	Lung	left lung (4404), right lung (4404), lung (3427)
	Lung lobes	left lung lower lobe (3454), right lung lower lobe (3454), right lung middle lobe (3454), left lung upper lobe (3454), right lung upper lobe (3454)
	Airways	trachea (2018), bronchi (583), airways (162)
	Ribs left	1st left rib (4108), 2nd left rib (4108), 3rd left rib (4108), 4th left rib (4108), 5th left rib (4108), 6th left rib (4108), 7th left rib (4108), 8th left rib (4108), 9th left rib (4108), 10th left rib (4108), 11th left rib (4108), 12th left rib (4108), 13th left rib (1597)
	Ribs right	1st right rib (4108), 2nd right rib (4108), 3rd right rib (4108), 4th right rib (4108), 5th right rib (4108), 6th right rib (4108), 7th right rib (4108), 8th right rib (4108), 9th right rib (4108), 10th right rib (4108), 11th right rib (4108), 12th right rib (4108), 13th right rib (1597), ribs (654)
	Sternum and clavicles	left clavicle (3974), right clavicle (3974), sternum (3454), rib cartilage (1761), sternum manubrium (533)
	Scapulae	left scapula (3974), right scapula (3974)
	Breast	left breast (1086), right breast (1086)
	Breast pathology	breast lesion (1506)
	Mediastinum	thymus (886), mediastinal tissue (533), mediastinal lymph node (393), mediastinal lymph nodes (192)
	Esophagus	esophagus (7549), cervical esophagus (84)
	Thoracic aorta	left subclavian artery (1931), brachiocephalic artery (1812), right subclavian artery (1761), subclavian artery (1)
	Thoracic veins	left brachiocephalic vein (1761), right brachiocephalic vein (1761), brachiocephalic veins (15), subclavian and carotid arteries (15), azygos (15), azygos vein (2), vertebral artery (1)
	Pleura	pleura lesion (245), pleural effusion (78)
	Lung pathology	lung lesion (1075), lung nodule (481), covid (209), lung tumor (113), non-small cell lung cancer (78), fdg-avid non-small cell lung cancer (69), fdg-avid lung adenocarcinoma (5)
Heart	Heart	heart (2855), heart tissue (533)
	Atria	left atrium (2516), right atrium (2386), left atrial appendage (1228)
	Ventricles	right ventricle (2800), left ventricular cavity (2503), left ventricle (703), right ventricular cavity (279)
	Atrial walls	left atrium wall (154), left atrium endocardium (154), atrium wall (10)
	Ventricular walls	left ventricle wall (3234)
	Myocardium	left ventricular myocardium (849), myocardium (700), excised myocardium (27)
	Valves	aortic valve (27), mitral valve (27)
	Coronary arteries	coronary artery (1000), left coronary artery (30), right coronary artery (30)
	Pulmonary circulation	pulmonary artery trunk (1713), pulmonary vein (1349), pulmonary artery (593), pulmonary artery tree (316), left atrium and pulmonary veins (137), pulmonary arterial tree (15)
	Systemic cardiac vessels	superior vena cava (1289), aorta with arch branches (110), aorta with branches and coronary arteries (27)
	Cardiac pathology	myocardial infarction (100), no reflow (100), left atrial scars (60), calcification in left anterior descending (32), calcification in left circumflex (32), calcification in right coronary artery (32)

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Region	Subregion	Structures (Count)
Abdomen	Liver parenchyma	liver (8669), liver without tumor or vessels (20)
	Liver segments	liver segment 1 (243), liver segment 2 (243), liver segment 3 (243), liver segment 5 (243), liver segment 6 (243), liver segment 7 (243), liver segment 8 (243), liver segment 4 (193), liver segment 4a (50), liver segment 4b (50)
	Liver pathology	liver lesion (4469), hepatic tumor (303), hepatocellular carcinoma (179), colorectal liver metastases (171), liver cancer (131), liver tumor (114), liver cyst (20)
	Pancreas	pancreas (7934), pancreas parenchyma (482), pancreatic duct (482)
	Pancreas pathology	pancreatic ductal adenocarcinoma lesion (482), pancreas tumor (423)
	Pancreas vessels	pancreatic veins (482), pancreatic arteries (482)
	Spleen	spleen (6823)
	Biliary system	gallbladder (5606), common bile duct (482), biliary system (20)
	Stomach and duodenum	stomach (5840), duodenum (4073)
	Small intestine	jejunum and ileum (3600), small bowel (50)
	Large intestine	colon (3506)
	Colon pathology	colon cancer primaries (126)
	Kidneys	left kidney (5666), right kidney (5603), kidney (3972)
	Kidney pathology	left kidney cyst (1228), right kidney cyst (1228), kidney tumor (559), kidney cyst (489), kidney masses (489)
	Adrenal glands	left adrenal gland (5423), right adrenal gland (5423), adrenal glands (140), left adrenal tumor (20), right adrenal tumor (20)
	Aorta abdominal	aorta (5057), descending aorta (135), aortic arch (135), abdominal aorta (120), abdominal aorta segment 1 (120), ascending aorta (segment) (120), aortic arch false lumen (120), abdominal aorta false lumen (120), abdominal aorta segment 1 false lumen (120), ascending aorta false lumen (120), aortic intimal tear (120), aorta vessel tree (56), thoracic aorta (30)
	Celiac axis	celiac trunk (533), splenic artery (121), celiac trunk / artery of the gut (120), celiac artery (50), coeliac trunk (1), splenic vein (1), left gastric artery (1), right gastric artery (1), short gastric veins (1), right gastroepiploic vein (1)
	Hepatic vessels	hepatic vessels (494), hepatic artery (120), common hepatic artery (1), gastroduodenal artery (1), hepatic artery proper (1), right hepatic artery (1), left hepatic artery (1), separate segment 5/6 vein (variant) (1), right hepatic vein (segment 7 only here, variant) (1), middle hepatic vein (1), left hepatic vein (1)
	Portal venous system	portal vein and splenic vein (2947), portal splenic vein (1597), portal veins (191), portal vein (21), venous system (20), right portal vein (1), left portal vein (1), jejunal branches of the superior mesenteric vein (1), ileal branches of the superior mesenteric vein (1), superior posterior pancreaticoduodenal vein (1), right colic vein (1), middle colic vein (1), pancreaticoduodenal veins (plexus) (1)
	Mesenteric vessels	superior mesenteric artery (171), inferior mesenteric artery (2), inferior mesenteric vein (2), superior mesenteric vein (1)
	Renal vessels	left renal artery (171), right renal artery (171), renal artery (70), renal vein (70), left renal vein (1), right renal vein (1)
	Inferior vena cava	inferior vena cava (5016)
	Gonadal vessels	left gonadal vein (2), right gonadal vein (1), a double/fenestrated right gonadal (1)
	Other abdominal vessels	left ascending lumbar vein (1), hemiazygos vein (1)
	Abdominal wall	abdominal tissue (533), fat (533)
	Other abdominal	stones (20), artery (20), metal (20), metastasectomy site (20)
Pelvis	Bladder	urinary bladder (5098), bladder (126)

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Region	Subregion	Structures (Count)
	Prostate	prostate (3003), prostate peripheral zone (269), prostate central gland (237), prostate transitional zone (32)
	Female reproductive	cervix (949), uterus (583), prostate and uterus (360)
	Gonads	gonads (886)
	Rectum	rectum (1281)
	Pelvic bones	sacrum (5720), left hip bone (5720), right hip bone (5720), left femoral head (604), right femoral head (604)
	Iliac vessels	left iliac artery (4514), right iliac artery (4514), left iliac vena (4514), right iliac vena (4514), right common iliac artery (172), left common iliac artery (172), right internal iliac artery (51), left internal iliac artery (51), right external iliac artery (51), left external iliac artery (51), right common iliac vein (2), left common iliac vein (1), right external iliac vein (1), left external iliac vein (1), right internal iliac vein (1), left internal iliac vein (1), right inferior epigastric artery (1), left inferior epigastric artery (1), right epigastric vein (1), left epigastric vein (1), right superficial epigastric vein(s) (1), left superficial epigastric vein(s) (1), middle sacral artery (1), middle sacral vein (1), right deep circumflex iliac artery (1), left deep circumflex iliac artery (1)
	Gluteal vessels	right superior gluteal vein (1), left superior gluteal vein (1), right inferior gluteal vein (1), left inferior gluteal vein (1), right inferior gluteal artery (1), left inferior gluteal artery (1), left superior gluteal artery (1)
	Femoral vessels	right femoral artery (121), left femoral artery (121), right common femoral artery (1), left common femoral artery (1), right common femoral vein (1), left common femoral vein (1), right femoral vein (1), left femoral vein (1), right deep femoral vein (1), right deep femoral artery (1), left deep femoral artery (1), right great saphenous vein (1), left great saphenous vein (1), left lateral circumflex vein (1), left lateral circumflex artery (1), right lateral circumflex artery (1), descending branch of the right lateral circumflex artery (1)
	Pelvic visceral vessels	right inferior vesical veins (1), left inferior vesical veins (1), right obturator veins (1), left obturator veins (1), right internal pudendal artery and vein (1), left internal pudendal artery and vein (1), right obturator artery (1), left obturator artery (1), right pubic veins (1), left ureter (1), right ureter (1), vesical plexus vein branch (only left side shown) (1)
	Rectal vessels	superior rectal artery (1), left colic artery (1), superior sigmoid artery (part of the sigmoid branches) (1), superior rectal vein (with branches) (1), sigmoid veins (1), left colic vein (1), left sacral veins (1), inferior rectal veins (1), right inferior rectal veins (1), left inferior rectal vein (1)
	Pelvic pathology	fdg-avid tumor (1159), psma-avid tumor (597), prostate tumor (188), clinically insignificant prostate lesion (140), clinically significant prostate lesion (140)
Spine	Cervical vertebrae	c1 vertebra (4504), c2 vertebra (4504), c3 vertebra (4504), c4 vertebra (4504), c5 vertebra (4504), c6 vertebra (4504), c7 vertebra (4504)
	Thoracic vertebrae	t9 vertebra (5546), t10 vertebra (5546), t11 vertebra (5546), t12 vertebra (5546), t7 vertebra (5099), t8 vertebra (5099), t1 vertebra (4559), t2 vertebra (4559), t3 vertebra (4559), t4 vertebra (4559), t5 vertebra (4559), t6 vertebra (4559), t13 vertebra (80)
	Lumbar vertebrae	l1 vertebra (5546), l2 vertebra (5546), l3 vertebra (5546), l4 vertebra (5546), l5 vertebra (5546), l6 vertebra (2743)
	Sacral	s1 vertebra (1228)
	Spinal cord and canal	spinal cord (5433), spinal canal (1383)

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Region	Subregion	Structures (Count)
	Intervertebral discs	intervertebral discs (616), l5 s1 disc (447), l4 l5 disc (447), l3 l4 disc (447), l2 l3 disc (447), l1 l2 disc (447), t12 l1 disc (447), t11 t12 disc (447), t10 t11 disc (447), t9 t10 disc (447), intervertebra disc (16)
	Other spine	vertebrae (616)
Extremities	Upper arm	left humerus (3974), right humerus (3974)
	Forearm	left radius (1597), right radius (1597), left ulna (1597), right ulna (1597)
	Hand	carpal left (1597), carpal right (1597), left metacarpal (1597), right metacarpal (1597), left fingers (1597), right fingers (1597)
	Femur	left femur (3974), right femur (3974), femur (100), left femur bone (78), right femur bone (78)
	Patella	left patella (1597), right patella (1597), left patella bone (78), right patella bone (78)
	Leg bones	left tibia (1597), right tibia (1597), left fibula (1597), right fibula (1597), tibia (100), left tibia bone (78), right tibia bone (78), left fibula bone (78), right fibula bone (78)
	Ankle bones	left tarsal (1597), right tarsal (1597), left calcaneous bone (78), right calcaneous bone (78), left talus bone (78), right talus bone (78)
	Foot bones	left metatarsal (1597), right metatarsal (1597), left toes (1597), right toes (1597), left cuboid bone (78), right cuboid bone (78), left intermediate cuneiform bone (78), right intermediate cuneiform bone (78), left lateral cuneiform bone (78), right lateral cuneiform bone (78), left medial cuneiform bone (78), right medial cuneiform bone (78), left navicular bone (78), right navicular bone (78), left phalanges bone (78), right phalanges bone (78)
	Pelvic bones	left pelvis bone (78), right pelvis bone (78)
	Femoral cartilage	femoral articular cartilage (100), left femur distal cartilage (78), right femur distal cartilage (78), left femur head cartilage (78), right femur head cartilage (78)
	Patellar cartilage	left patella cartilage (78), right patella cartilage (78)
	Tibial cartilage	tibial articular cartilage (100), left tibia distal cartilage (78), right tibia distal cartilage (78), left tibia lateral cartilage (78), right tibia lateral cartilage (78), left tibia medial cartilage (78), right tibia medial cartilage (78)
	Other cartilage	left pelvis acetabulum cartilage (78), right pelvis acetabulum cartilage (78), left talus cartilage (78), right talus cartilage (78)
	Knee ligaments	left acl ligament (78), right acl ligament (78), left lcl ligament (78), right lcl ligament (78), left mcl ligament (78), right mcl ligament (78), left pcl ligament (78), right pcl ligament (78)
Musculo-skeletal	Paraspinal muscles	left autochthon (4514), right autochthon (4514), left longissimus thoracis (97), right longissimus thoracis (97), right spinalis thoracis (1), left spinalis thoracis (1), right ilicostalis lumborum (1), left iliocostalis lumborum (1)
	Abdominal muscles	right quadratus lumborum (1), left quadratus lumborum (1), right rectus abdominis (1), left rectus abdominis (1), right transversus abdominis (1), left transversus abdominis (1), right external oblique (1), left external oblique (1), right internal oblique (1), left internal oblique (1)
	Gluteal muscles	left gluteus maximus (4611), right gluteus maximus (4611), left gluteus medius (4611), right gluteus medius (4611), left gluteus minimus (4071), right gluteus minimus (4071), left gluteus maximus muscle (78), right gluteus maximus muscle (78), left gluteus medius muscle (78), right gluteus medius muscle (78), left gluteus minimus muscle (78), right gluteus minimus muscle (78)

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Region	Subregion	Structures (Count)
	Hip flexors	left iliopsoas (4610), right iliopsoas (4610), right psoas major muscle (78), left psoas major muscle (78), right iliacus muscle (78), left iliacus muscle (78), right psoas (1), left psoas (1), right iliacus (1), left iliacus (1)
	Hip rotators	right piriformis muscle (78), left piriformis muscle (78), right superior gemellus muscle (78), left superior gemellus muscle (78), right inferior gemellus muscle (78), left inferior gemellus muscle (78), right obturator internus muscle (78), left obturator internus muscle (78), right obturator externus muscle (78), left obturator externus muscle (78), left quadratus femoris muscle (78), right quadratus femoris muscle (78), right piriformis (1), left piriformis (1), right superior gemellus (1), left gemellus superior (1), right inferior gemellus (1), left gemellus inferior (1), right obturator internus (1), left obturator internus (1), right obturator externus (1), left obturator externus (1), right quadratus femoris (1)
	Hip abductors	right tensor fasciae latae muscle (78), left tensor fasciae latae muscle (78), right tensor fascia lata (1), left tensor fascia lata (1)
	Pelvic floor	right ischiocavernosus (1), left ischiocavernosus (1)
	Quadriceps	right rectus femoris muscle (78), left rectus femoris muscle (78), right vastus intermedius muscle (78), left vastus intermedius muscle (78), right vastus lateralis muscle (78), left vastus lateralis muscle (78), left vastus medialis muscle (78), right vastus medialis muscle (78), right rectus femoris (1), left rectus femoris (1), right vastus intermedius (1), left vastus intermedius (1), right vastus lateralis (1), left vastus lateralis (1)
	Other anterior thigh	right sartorius muscle (78), left sartorius muscle (78), right sartorius (1), left sartorius (1)
	Adductors	right pectineus muscle (78), left pectineus muscle (78), right gracilis muscle (78), left gracilis muscle (78), right adductor longus muscle (78), left adductor longus muscle (78), right adductor brevis muscle (78), left adductor brevis muscle (78), right adductor magnus muscle (78), left adductor magnus muscle (78), right pectineus (1), left pectineus (1), right gracilis (1), left gracilis (1), right adductor longus (1), left adductor longus (1), right adductor brevis (1), left adductor brevis (1), right adductor minimus (1), left adductor minimus (1), right adductor magnus (1), left adductor magnus (1)
	Hamstrings	right biceps femoris long head muscle (78), left biceps femoris long head muscle (78), right biceps femoris short head muscle (78), left biceps femoris short head muscle (78), right semitendinosus muscle (78), left semitendinosus muscle (78), right semimembranosus muscle (78), left semimembranosus muscle (78), right biceps femoris (long head) (1), left biceps femoris (long head) (1), right semitendinosus (1), left semitendinosus (1), right semimembranosus (1), left semimembranosus (1)
	Calf posterior	left gastrocnemius lateral muscle (78), right gastrocnemius lateral muscle (78), left gastrocnemius medial muscle (78), right gastrocnemius medial muscle (78), left soleus muscle (78), right soleus muscle (78), left plantaris muscle (78), right plantaris muscle (78), left popliteus muscle (78), right popliteus muscle (78)
	Calf anterior	left tibialis anterior muscle (78), right tibialis anterior muscle (78), left extensor digitorum longus muscle (78), right extensor digitorum longus muscle (78), left extensor hallucis longus muscle (78), right extensor hallucis longus muscle (78)
	Calf deep posterior	left tibialis posterior muscle (78), right tibialis posterior muscle (78), left flexor digitorum longus muscle (78), right flexor digitorum longus muscle (78), left flexor hallucis longus muscle (78), right flexor hallucis longus muscle (78)
	Calf lateral	left peroneus longus muscle (78), right peroneus longus muscle (78)

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Region	Subregion	Structures (Count)
	General musculoskeletal	bones (2070), skin (886), muscles (533)
Vascular	Aorta and branches	aortic arch (135), descending aorta (135), abdominal aorta (120), innominate artery (120), innominate artery segment 1 (120), innominate artery segment 1 false lumen (120), right common iliac artery false lumen (120), left common iliac artery false lumen (120), left common carotid artery false lumen (120), left subclavian artery false lumen (120), descending aorta false lumen (120), ascending aorta (101), aorta zone 0 (50), aorta zone 1 (50), aorta zone 2 (50), aorta zone 3 (50), aorta zone 4 (50), aorta zone 5 (50), aorta zone 6 (50), aorta zone 7 (50), aorta zone 8 (50), aorta zone 9 (50), thoracic aorta (30)
	Venous system	inferior vena cava (5016), superior vena cava (1289), portal vein (21), azygos vein (2), splenic vein (1)
	Vascular pathology	true lumen (100), false lumen (100), false lumen thrombus (100), false aortic lumen (39), true aortic lumen (39), pulmonary embolism (35)
General pathology	Tumors and lesions unspecified	lesion (1093), malignant melanoma tumor (670), fdg-avid soft tissue sarcoma (42), tumor (unspecified) (20)

Table 10. **Summary of anatomical structures.** Overview of all 1078 distinct anatomical labels from public training datasets. Each entry lists the structure name and its training case frequency (in parentheses), organized by anatomical region and subregion.